Project Plan – Graduation Internship

Quantum Imaging with Undetected Photons: Enhanced Image processing & User Interface

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Institution:	Saxion University of Applied Sciences

Internship period: 02.02.2026 – 03.07.2026

Contents

1 Introduction **4**

1.1 Background Information 4

1.2 The Setup 4

2 Assignment Description **7**

2.1 Objectives 7

2.2 Project Requirements 7

2.3 Scope of Work 8

3 Organization **9**

4 Planning **10**

5 Methodology **11**

5.1 Quality Control 11

5.2 Development Tools 12

5.3 Communication and Reporting 12

5.4 Approach & Strategy 12

5.5 FFT-Based Image Processing 13

6 Risk Analysis & Achievability **15**

6.1 Risk Analysis 15

6.2 Achievability 15

7 Appendix **16**

A Gantt Chart **16**

List of Figures

1 Demonstrational Setup and Schematic. 5

2 The Scrum Cycle 11

3 Planning Timeline – Gantt Chart 16

List of Abbreviations

GUI Graphical User Interface

CMOS Complementary Metal-Oxide-Semiconductor

ANT Applied Nanotechnology

QIUP Quantum Imaging with Undetected Photons

PPLN Periodically-Poled Lithium Niobate

SPDC Spontaneous Parametric Down-Conversion

FFT Fast Fourier Transform

IDE Integrated Development Environment

SDK Software Development Kit

DFT Discrete Fourier Transform

CCD Charged Coupled Device

1 Introduction

This graduation project is conducted from February 2nd, 2026, to July 3rd, 2026, as part of the Bachelor of Applied Computer Science program at Saxion University of Applied Sciences. The primary objective is to develop a robust automation framework and an intuitive Graphical User Interface (GUI) for the quantum imaging demonstrator shown in Figure 1. By integrating hardware control with image processing, this project aims to enhance the accessibility and engagement with quantum optical phenomena for educational purposes.

This Project Plan serves as a roadmap for the internship, detailing the technical approach, project management methodology, and expected outcomes.

1.1 Background Information

This project is part of the Quantum Delta program [1], which aims to accelerate quantum technology developments by fostering a collaborative ecosystem.

A primary challenge in this field is the non-intuitive nature of quantum properties such as entanglement and indistinguishability. The demonstrator addresses this by utilizing Quantum Imaging with Undetected Photons (QIUP) [2].

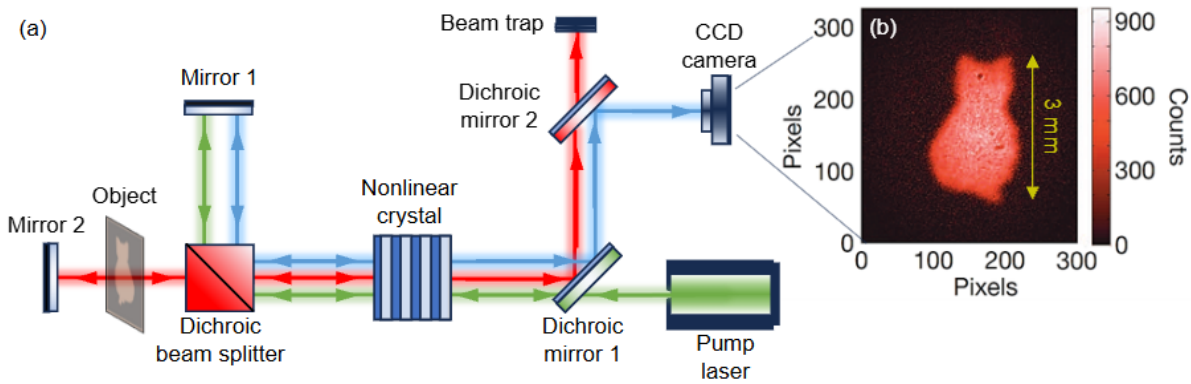
1.2 The Setup

The experimental implementation of the quantum imager is shown in Figure 1. The system consists of:

- Continuous-wave pump laser
- Periodically-Poled Lithium Niobate (PPLN) nonlinear crystal
- Dichroic beam splitter
- Piezoelectric-stage connected with a mirror
- Thorlabs Zelux Complementary Metal-Oxide-Semiconductor (CMOS) camera

Inside the PPLN crystal, Spontaneous Parametric Down-Conversion (SPDC) splits pump photons into signal (visible) and idler (infrared) photons. The idler interacts with the object, while the signal does not. Upon recombination, interference occurs and is recorded by the CMOS camera.

(a) Top-Down view on demonstrational setup at Saxion.



(b) Schematic of demonstrational setups functionality.

Figure 1: Demonstrational Setup and Schematic.

The system relies on the principle of induced coherence, utilizing a nonlinear interferometer.

As detailed in the schematic (Figure 1b), a continuous wave pump laser is directed into a nonlinear crystal specifically, a PPLN crystal. Through SPDC, the pump photon splits into a signal photon (visible spectrum) and an idler photon (infrared). A dichroic beam splitter separates these paths: the visible signal is reflected by a reference mirror (Mirror 1), while the infrared idler interacts with the target object before being reflected by Mirror 2.

The idler photons that pass through the object contain spatial information about its absorption or phase profile. When the idler and signal beams recombine at the crystal, quantum interference modulates the intensity of the signal beam based on the idler's path information. Consequently, the image of the object which was only "seen" by the infrared light is transferred to the visible spectrum and detected by the Charged Coupled Device (CCD) camera.

The physical realization of this schematic is shown in Figure 1a. The setup is constructed on an optical breadboard to ensure stability. The pump source directs the beam through conditioning optics towards the central crystal mount. The detection module utilizes a Thorlabs Zelux CMOS camera to capture the interference patterns.

2 Assignment Description

The assignment focuses on transforming a manually controlled setup into a robust, automated, and user-friendly demonstrator for QIUP. Currently, the system relies on generic camera software without real-time image processing.

2.1 Objectives

The main objective is to bridge the gap between the physical optical setup and the end-user by developing an integrated software solution. This involves synchronizing hardware control, implementing on-the-fly data processing, and wrapping these functionalities into an accessible interface for educational demonstrations.

2.2 Project Requirements

To consider this project successful, the expected technical results have been categorized using the MoSCoW method:

Must Have:

- Hardware Automation System: A robust Python-based control script capable of communicating with and reliably driving both the Thorlabs CMOS camera and the piezoelectric stage.
- Image Processing Core: A functional Fast Fourier Transform (FFT)-based algorithm capable of real-time or near real-time reconstruction of visibility, contrast, and phase images from the raw sensor data.

Should Have:

- Functional GUI: An intuitive graphical user interface that allows the operator to control the hardware, acquire live images, and display the reconstructed quantum interference patterns.
- Performance Optimization: Validated and optimized code execution that ensures the imaging performance is fast and smooth enough for live audience demonstrations.

Could Have:

- Advanced GUI Capabilities: Additional features in the software, such as one-click calibration presets, advanced data visualization tweaks, or the ability to export acquired data.

Won't Have:

- Hardware Upgrades: The physical integration or programming of new hardware components not present in the current setup.

Deliverables:

- Project plan.
- Python automation software.
- Test results.
- Technical and operational documentation.
- Final thesis report.

2.3 Scope of Work

The scope is limited to software automation and image processing. No hardware modification will be done and it will fully rely on the currently available setup.

The primary focus will be set on the image processing to retrieve the desired pictures with different numbers of frames for testing purposes.

3 Organization

This section shows the general organizational aspects of this project discussing all stakeholders and their contribution.

Supervisor & Study Coach

The project is supervised by Dr. D. Polishchuk, a senior researcher at Applied Nanotechnology (ANT). Weekly meetings ensure technical alignment and progress monitoring.

Dr. A. Khan from the Applied Computer Science program will be the study coach.

Stakeholders

The success of this demonstrator project is supported by a collaboration between several key organizations within the Dutch quantum ecosystem.

- ANT Research Group: As the primary host of the internship, the ANT group at Saxion University of Applied Sciences is responsible for the physical development, image processing and scientific validation of the quantum setup.
- Quantum Delta NL: With the goal to constantly create, improve or innovate the dutch quantum infrastructure they are partners in research and educational purposes with QIUP.

Workspace

The project activities will be divided between two primary locations at Saxion, depending on the tasks:

- Experimental Laboratory: Hardware integration, troubleshooting, and system testing will be conducted in the TLC Lab (Room Z.14.0.02) at the Ariënsplein campus, where the physical quantum imaging demonstrator is.
- Office Workspace: Technical documentation will primarily take place in the OTSWO area (W3) of the Saxion Main Building.

Additionally, homeoffice arrangements for tasks that do not require physical access to the hardware setup can be discussed in agreement with the project supervisor.

4 Planning

The project can be divided into 3 main tasks:

- Documentation: Code quality, user manuals, and thesis writing.
- Research: Quantum optics literature, FFT theory, and hardware Software Development Kit (SDK) analysis.
- Implementation: Python drivers, acquisition synchronization, and GUI development.

A Gantt chart visualizes a rough project timeline setting milestones and development sprints which is provided in Appendix A.

5 Methodology

To manage the iterative nature of software development alongside research and hardware integration, this project utilizes Scrum. The Agile approach allows for flexibility in development while ensuring continuous delivery of software features. Figure 2 demonstrates how a Scrum cycle works, visualizing the advantage for iterative progress.

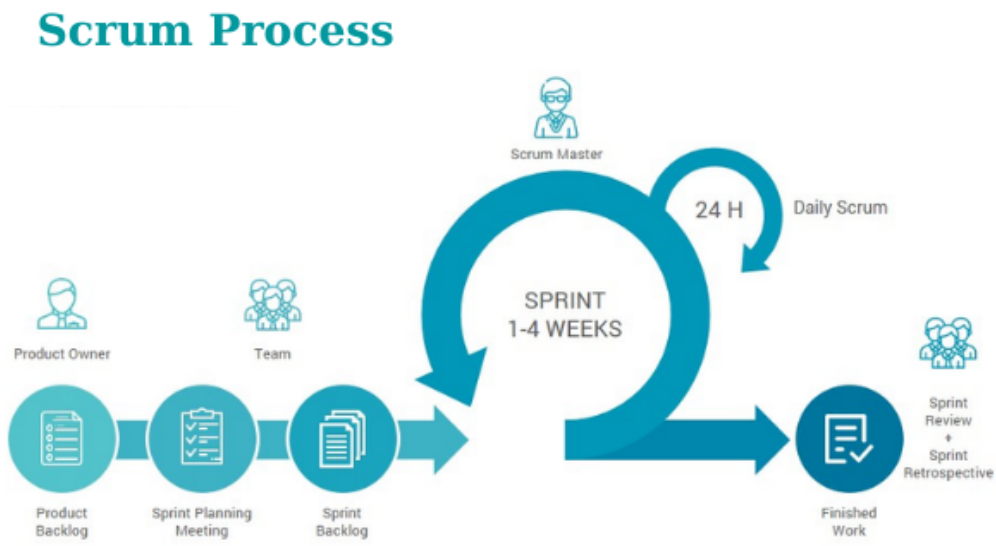


Figure 2: The Scrum Cycle

The project is divided into 1-4 weeks sprints. This duration provides a balance between sufficient development time and frequent feedback loops. At the end of each sprint the past sprint will be reviewed and the next sprint will be adjusted according to past progress. Figure 2 shows the cycle for each sprint.

5.1 Quality Control

To maintain code maintainability and system stability, a development sprint or task is only considered complete when it meets the following criteria:

- The code is written in Python and properly commented.
- The feature is version-controlled and pushed to the main Git repository.
- The functionality has been successfully tested and validated on the physical quantum imaging setup.

5.2 Development Tools

- Version Control: Git and GitHub will be used for source code management.
- Integrated Development Environment (IDE): Visual Studio Code is the chosen IDE for Python development.
- Scrum: ClickUp will be used to keep track of all sprints and their corresponding subtasks.
- Documentation: LaTeX will be used for all technical documentation and deliverables to ensure professional formatting and visual consistency.
- Python: Python will be the Programming language used to program the entire system.

5.3 Communication and Reporting

To maintain project momentum, the following communication protocols are established:

- Weekly Stand-ups: Formal progress review every week to discuss Sprint outcomes and address challenges.
- Meeting Documentation: All meeting notes will be published on GitHub.

5.4 Approach & Strategy

To achieve the main objective of creating a fully automated GUI capable of live quantum image reconstruction, the most promising analysis technique is FFT. The implementation will rely on the FFTW library [3], widely regarded as the industry standard for performance. While some libraries approximate continuous sinusoids, FFTW leverages the Discrete Fourier Transform (DFT) to provide a more efficient framework for digital filtering. This discrete approach is better for calculating the frequency-domain adjustments required to enhance visibility, contrast, and phase imaging.

To surpass the signal-to-noise limitations of standard single-shot imaging, the experimental setup will use a piezoelectric stage. This device, mounted on the reference mirror, enables precise control over the optical path length of the interferometer.

5.5 FFT-Based Image Processing

Converting Pixels to Frequencies

Most images are stored in the spatial domain, where we see pixel brightness as a function of position. However, for an interference-based system like the QIUP demonstrator, the most important data is encoded in the patterns (fringes) of the light. By using a FFT, we shift the image into the frequency domain.

In this frequency-based view, the image is broken down by detail rather than position:

- Low Frequencies: Represent the general background lighting and smooth intensity variations.
- High Frequencies: Represent sharp edges, fine details, and random camera noise.

By isolating specific frequencies, we can "tune out" background noise and reconstruct a much clearer image focusing on the sample's phase and contrast.

Why pyFFTW Over Standard Methods?

While standard libraries like numpy.fft are common, they often use a "one-size-fits-all" approach. For a real-time demonstrator where every millisecond counts, we use pyFFTW, the Python interface for the "Fastest Fourier Transform in the West" library.

The primary advantage is the Execution Planner. Instead of running a generic formula, pyFFTW treats the FFT as an optimization problem tailored to your specific computer hardware.

Feature	**NumPy / Standard FFT**	**pyFFTW (Our Choice)**
Strategy	Uses a fixed, general algorithm.	Plans the best path for your CPU.
Memory	Standard allocation.	Aligned memory for faster data pipes.
Speed	Good for one-off calculations.	Optimized for repeated frames.

Table 1: Comparison between standard FFT and pyFFTW.

The Planning Phase: How it Self-Optimizes

Before the live camera feed begins, pyFFTW performs a "warm-up" known as Planning. During this stage, the library tests several different mathematical strategies to see which one runs fastest on the specific processor. It evaluates a variety of classic and modern FFT algorithms, including:

- Cooley-Tukey: The standard approach that breaks large problems into smaller, recursive pieces.
- Split-Radix: A specialized version that minimizes the number of actual multiplications required.
- Bluestein & Rader: These allow the system to maintain high speeds even if the image dimensions are not a perfect power of two.
- SIMD Vectorization: Methods that ensure the CPU processes multiple data points simultaneously.

Real-Time Benefits

Once the optimal "Plan" is found, it is saved (as Wisdom) and reused for every single frame coming off the camera. This reduces the time spent calculating each frame, resulting in:

- Higher Frame Rates: A smoother "live" video feel for the user.
- Lower Latency: Minimal delay between the sample moving and the processed image updating on screen.
- Efficiency: The CPU stays cooler and has more headroom for other tasks, such as phase reconstruction or data saving.

6 Risk Analysis & Achievability

This section identifies the primary technical risks and the strategy to ensure project completion within the five-month period.

6.1 Risk Analysis

Risk	Description	Mitigation Strategy
Hardware Interfacing	Camera or piezo stage SDK may lack native Python 3 support.	Use Python wrappers to interface with existing C-based drivers.
Phase Artifacts	DFT calculations may produce phase wrapping (cycle jumps) or aliasing.	Calibrate piezo step size and implement basic phase-unwrapping algorithms.
Performance	Real-time constraints	Optimize FFT planning

Table 2: Primary Project Risks

6.2 Achievability

Despite the technical risks, this project is highly achievable. The most significant advantage is that the optical setup is already built, calibrated, and physically present, eliminating any delays related to ordering or assembling parts. With the hardware ready and expert supervision available for the physics aspects, the project goals are realistic and attainable within the five-month timeframe.

7 Appendix

A Gantt Chart

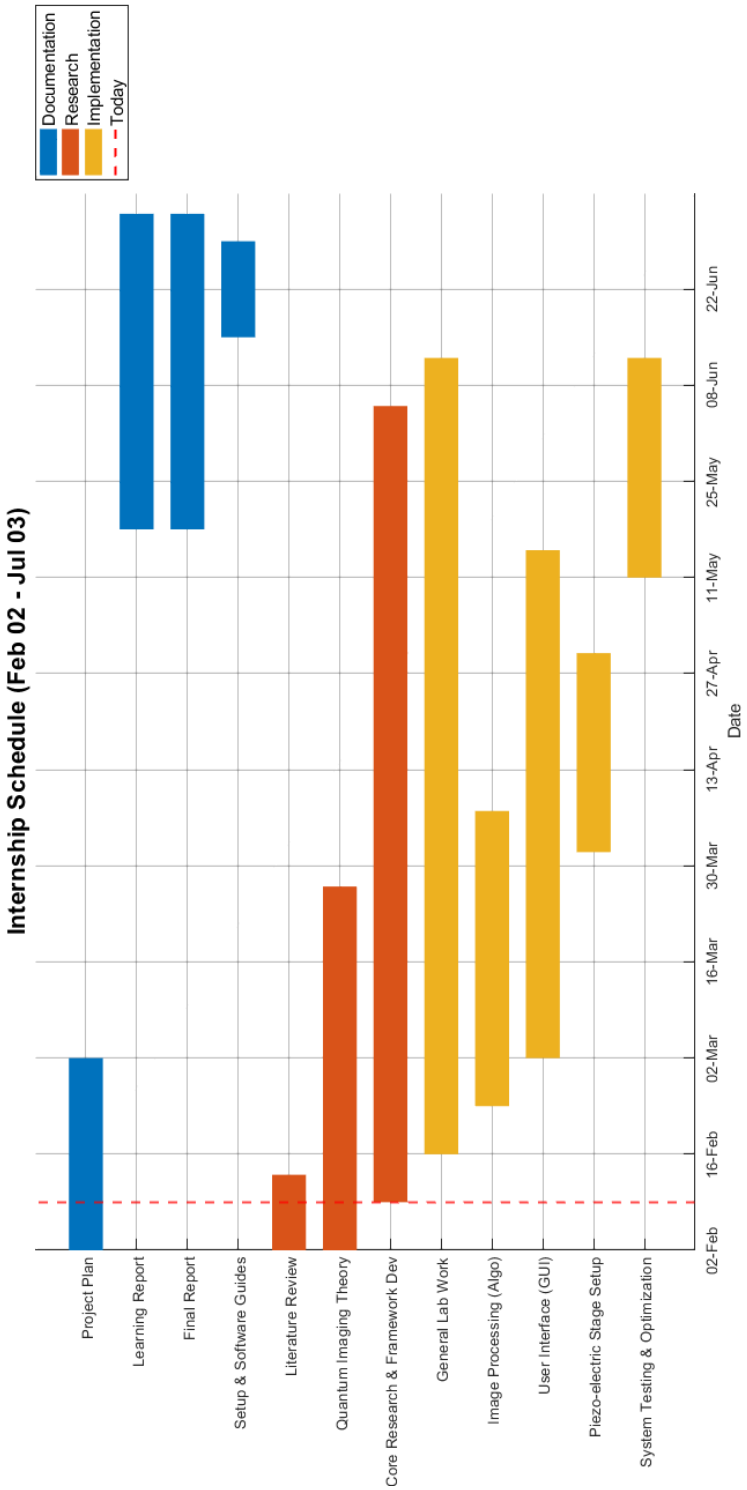


Figure 3: Planning Timeline – Gantt Chart

References

[1] Quantum Delta NL, "Quantum delta NL," 2024.

[2] G. B. Lemos, V. Borish, G. D. Cole, S. Ramelow, R. Lapkiewicz, and A. Zeilinger, "Quantum imaging with undetected photons," *Nature*, vol. 512, no. 7515, pp. 409–412, 2014.

[3] M. Frigo and S. G. Johnson, "The design and implementation of fftw3," *Proceedings of the IEEE*, vol. 93, no. 2, pp. 216–231, 2005.